

■ GENERAL DATA ANALYST Interview Questions & Answers

For Freshers — HR, Statistics, Scenario & Behavioural Questions

50+ Questions covering Statistics, Scenario, HR and Fresher-specific questions

SECTION 1 — HR & Introduction Questions

Q1. Tell me about yourself.

Answer: Give a 2-minute structured answer: 1) Education background. 2) Skills you've learned (SQL, Python, Excel, Power BI). 3) Projects or internships. 4) Why you want to be a Data Analyst. Keep it professional and relevant.

Example: I am a recent graduate in [field] from [college]. During my studies, I learned SQL, Python, and Excel, and worked on a project where I analysed [topic] using [tools]. I am passionate about turning data into insights and looking forward to starting my career as a data analyst.

Q2. Why do you want to become a Data Analyst?

Answer: Talk about genuine interest in data and problem-solving. Mention specific skills you enjoy (pattern finding, making sense of numbers, creating visuals). Relate it to business impact — how data helps companies make better decisions.

Remember: Be honest and specific. Don't just say 'I like working with data' — say WHY and WHAT specifically excites you.

Q3. What are your strengths for a Data Analyst role?

Answer: Choose 2-3 real strengths and give an example for each: Attention to detail (important for data quality), Problem-solving mindset, Quick learner with tools, Good at storytelling with data (explaining insights simply), Strong Excel/SQL skills.

Q4. What is your weakness? (The honest answer interviewers like)

Answer: Choose a real weakness that is NOT critical, and show how you are WORKING ON IT. Example: 'I sometimes spend too much time making things perfect — I am learning to set time limits for tasks and focus on the most important insights first.'

Remember: Never say 'I have no weaknesses' or 'I work too hard' — interviewers don't like fake answers.

Q5. Where do you see yourself in 3-5 years?

Answer: Show ambition but keep it realistic. Example: 'In 3 years, I see myself as a confident Data Analyst who can independently handle end-to-end analysis projects. In 5 years, I would like to move towards Senior Analyst or Data Science roles.'

Q6. Why should we hire you?

Answer: Connect your skills directly to the job requirements. Example: 'I have hands-on knowledge of SQL, Python, Excel, and Power BI. I am a quick learner, detail-oriented, and I can communicate data insights in simple terms. I am motivated and willing to work hard to contribute to your team from day one.'

Q7. Do you have any projects or experience?

Answer: For freshers — mention academic projects, personal projects, internships, or online courses with practical work. Describe: What problem you solved, what data you used, what tools you used, and what insight you found. Even a small self-practice project counts!

Example: I analysed an e-commerce dataset from Kaggle using Python and Excel. I cleaned the data, found that 40% of sales came from 3 cities, and created a dashboard. This project helped me learn Pandas and Pivot Tables.

SECTION 2 — Statistics & Concepts (Easy Questions)

Q8. What is data analytics in simple words?

Answer: Data analytics means collecting raw data, cleaning it, analysing it, and finding useful information (insights) that helps make better business decisions. It's like being a detective — you look at the clues (data) and figure out what's happening and why.

Q9. What are the 4 types of analytics?

Answer: Descriptive — WHAT happened? (Reports, dashboards, summaries of past data). Diagnostic — WHY did it happen? (Root cause analysis, drill-down). Predictive — WHAT WILL happen? (Forecasting using past patterns). Prescriptive — WHAT SHOULD WE DO? (Recommendations and optimisation).

Remember: Descriptive→Diagnostic→Predictive→Prescriptive — each builds on the previous.

Q10. What is KPI?

Answer: KPI = Key Performance Indicator. It is a measurable value that shows how well a business is achieving its goals. Examples: Monthly Revenue, Number of New Customers, Customer Satisfaction Score (CSAT), Website Conversion Rate, Employee Attrition Rate.

Remember: KPIs are what businesses measure to track success.

Q11. What is the difference between quantitative and qualitative data?

Answer: Quantitative data = numbers you can measure and calculate. Examples: age, salary, temperature, sales amount. Qualitative data = categories or descriptions, not numbers. Examples: gender, city, color, product category, feedback text.

Remember: Quantitative = numbers. Qualitative = categories/descriptions.

Q12. What is the difference between structured and unstructured data?

Answer: Structured data = organised in rows and columns (like a database or Excel file). Easy to analyse. Unstructured data = no fixed format — images, videos, emails, social media posts, PDFs. Harder to analyse, needs special tools (NLP, computer vision).

Remember: Most SQL and Excel work = structured data.

Q13. What is a KPI dashboard?

Answer: A KPI dashboard is a single visual screen that shows all the important business metrics at a glance. Like a car dashboard — one look tells you speed, fuel, temperature. Business dashboards show: Sales, Revenue, Growth %, Customer Count, etc. Built using Power BI, Excel, or Tableau.

Q14. What is data-driven decision making?

Answer: Making decisions based on actual data and analysis rather than gut feeling or guesswork. Example: instead of guessing which product to promote, analyse sales data to see which product has the highest growth rate and promote that one.

Q15. What is the difference between a database and a spreadsheet?

Answer: Spreadsheet (Excel) — small to medium data, one user at a time, good for analysis and reporting, easy to use. Database (SQL) — large amounts of data, multiple users simultaneously, structured with relationships between tables, more secure and reliable for organisations.

SECTION 3 — Statistics Interview Questions

Q16. What is the difference between Mean, Median, and Mode?

Answer: Mean = add all values / count of values = average. Median = middle value when sorted. Mode = most frequently appearing value. When to use: Mean for symmetric data. Median when there are outliers (house prices, salaries). Mode for categorical data.

Example: Salaries: 25K, 28K, 30K, 30K, 200K. Mean = 62.6K (misleading). Median = 30K (better). Mode = 30K.

Q17. What is Standard Deviation in simple words?

Answer: Standard deviation tells you how much the values in your data differ from the average. If SD is small = values are close together. If SD is large = values are spread out widely.

Example: Test marks: 85,87,86,85,88 — low SD (consistent class). Marks: 30,60,95,40,100 — high SD (inconsistent class).

Q18. What is a p-value? (Simple explanation for freshers)

Answer: A p-value tells you if your result is statistically significant or just happened by luck. Rule: If p-value < 0.05, the result is significant (not by chance). If p-value >= 0.05, the result might be due to chance.

Example: You tested a new ad. Sales increased. p-value = 0.03 → the increase is real, not by chance.

Remember: $p < 0.05$ = significant result. $p \geq 0.05$ = might be random.

Q19. What is A/B Testing in simple words?

Answer: A/B testing is comparing two versions of something to see which one performs better. Group A gets the original version. Group B gets the new version. You measure and compare the results using statistics.

Example: Website has two button colors: Blue (A) vs Green (B). Which gets more clicks? Test on real users and measure.

Q20. What is correlation? Give an example.

Answer: Correlation shows the relationship between two variables — if one changes, does the other change too? Positive correlation: both increase together (height and weight). Negative correlation: one increases, other decreases (exercise and weight). Zero correlation: no relationship.

Remember: Correlation does NOT mean one causes the other!

Q21. What is a sample and a population?

Answer: Population = the entire group you want to study. Example: all customers of a company. Sample = a small subset of the population that you actually study. You use sample data to make conclusions about the whole population.

Example: You can't survey all 10 million customers — survey 1000 (sample) and draw conclusions.

Q22. What is Hypothesis Testing in simple words?

Answer: Hypothesis testing is a way to check if an assumption about data is true or false. You start with a NULL HYPOTHESIS (H_0 = no change, no difference). Then you check data to see if there's enough evidence to REJECT the null hypothesis.

Example: H_0 : New feature doesn't improve sales. You test it. If data shows significant improvement ($p < 0.05$), reject H_0 — the feature works!

SECTION 4 — Scenario & Practical Questions

Q23. How do you approach a new data analysis project?

Answer: Step 1: Understand the business problem — what question are we answering? Step 2: Identify what data is available. Step 3: Collect and load the data. Step 4: Clean the data (handle nulls, duplicates, wrong types). Step 5: Explore the data (EDA). Step 6: Analyse — find patterns, trends, answers. Step 7: Visualise — create charts/dashboards. Step 8: Present insights with recommendations.

Remember: Always start with the business question — data work without a clear goal is wasted effort.

Q24. Your data has 30% missing values in one column. What do you do?

Answer: First, understand WHY values are missing (is it random or systematic?). If missing randomly and column is not critical — might drop the column. If column is important — try imputing (fill with mean/median/mode). If pattern exists in missing (MNAR) — keep a flag column noting it was missing. Always document what you did and why.

Q25. You find an outlier in your data. What steps do you take?

Answer: Step 1: Identify it (box plot, IQR method). Step 2: Investigate — is it a data error (typo, wrong entry) or a real extreme value? Step 3: If error — fix or remove it. If real — decide based on analysis goal: keep it (if studying extremes), cap it (winsorise), or transform data (log scale). Always document your decision.

Q26. A stakeholder says 'sales are down this month'. How do you investigate?

Answer: 1) Pull sales data and confirm the drop is real. 2) Compare to same period last year (seasonal?). 3) Break down by product, region, customer type — where exactly did it drop? 4) Look at external factors (competition, holidays, price changes). 5) Check if there were data collection issues. 6) Present findings with visualisations.

Remember: This is the Diagnostic Analytics approach — always break the problem down.

Q27. How do you explain technical findings to a non-technical manager?

Answer: 1) Start with the KEY INSIGHT first (the answer), not how you got there. 2) Use simple words — no jargon. 3) Use visuals — charts are universal. 4) Use analogies to make it relatable. 5) Focus on BUSINESS IMPACT — what does this mean for revenue/cost/customers?

Example: Instead of 'The correlation coefficient is 0.87' say 'Cities with more stores almost always have higher sales — they go hand in hand.'

Q28. What would you do if your analysis gives a result that contradicts what the business team believes?

Answer: 1) Double-check your analysis for mistakes first. 2) If analysis is correct, present the findings transparently with clear evidence. 3) Be respectful — they may have context you don't. 4) Explore their assumptions — maybe data tells a different story. 5) Let the data lead, but stay open to discussion.

Remember: Data analysts present findings honestly — not what people want to hear.

Q29. What tools would you use for a full data analysis project?

Answer: Data storage: SQL database or Excel. Data cleaning: Python (Pandas) or Excel Power Query. Analysis: Python or SQL queries. Visualisation: Power BI or Excel charts. Reporting: PowerPoint or PDF. Tools depend on data size and company tools available.

SECTION 5 — Fresher-Specific Tips & Final Questions

Q30. What projects can a fresher do to build a portfolio?

Answer: 1) Download free datasets from Kaggle.com, data.gov.in, or UCI ML Repository. 2) Do EDA on any dataset (sales data, Titanic, IPL, movies). 3) Clean a messy real-world dataset and document every step. 4) Build a Power BI or Excel dashboard. 5) Write a summary of your findings. Projects show interviewers you know HOW to do the work.

Example: Kaggle has hundreds of free datasets. Start with 'Superstore Sales', 'IPL matches', or 'Covid data'.

Q31. What are common mistakes freshers make in interviews?

Answer: 1) Not practicing writing actual SQL queries/Python code — just reading theory is not enough. 2) Not having a project to show. 3) Saying 'I know Excel' without knowing Pivot Tables and VLOOKUP. 4) Not being able to explain concepts in simple words. 5) Forgetting to ask about the company's tech stack.

Q32. Do you have any questions for us? (Always say YES!)

Answer: Good questions to ask: 'What tools does the team currently use for analysis?' / 'What would a typical day look like for this role?' / 'What are the main data sources I would work with?' / 'What does success look like in the first 90 days?' Asking good questions shows genuine interest.

Remember: Always prepare 2-3 questions — never say 'No, I don't have any questions'.

Q33. How do you stay updated with data analytics trends?

Answer: Mention: Following data-related blogs (Towards Data Science, Analytics Vidhya). LinkedIn learning. Kaggle competitions. YouTube tutorials. Taking free courses on Coursera, Udemy, Google Data Analytics Certificate. Practicing on real datasets regularly.

Q34. What is the difference between Data Analyst, Data Scientist, and Data Engineer?

Answer: Data Analyst — uses SQL, Excel, Python to analyse EXISTING data and find insights. Focuses on PAST data. Data Scientist — builds ML models to PREDICT future outcomes. More advanced maths/statistics. Data Engineer — builds and maintains the PIPELINES and infrastructure that stores and delivers data. More programming and cloud focus.

Remember: As a fresher, be clear you're targeting Data Analyst role and know what it involves.

Interview Day Checklist & Tips

Before Interview	During Interview	After Interview
Revise SQL queries	Speak slowly and clearly	Send a thank-you email
Practice Pivot Tables	Think aloud — explain your approach	Reflect on what went well/didn't
Review your projects	If unsure, say 'Let me think...'	Practice questions you couldn't answer
Research the company	Use examples from your projects	Keep applying — don't give up!
Prepare 3 questions to ask	Ask for clarification if needed	Update your resume if needed
Good sleep the night before	Show enthusiasm for data	Follow up after 3-5 days
Carry printed resume	Don't panic on tough questions	Learn from every interview

YOU GOT THIS! ■

Every expert was once a beginner. Keep practicing, keep learning, and your first job is just around the corner!